Japanese consolidated balance-sheet parser: project report

PDF page (text layer or scan) in, verified Excel table out. Personal, non-commercial project. Code and test material: github.com/premsuggu/japanese-financial-table-parser. October 2026.

Goal

Read a page of a Japanese consolidated balance sheet whose layout may vary and which may be scanned, and produce a correct table: row labels, values, columns, hierarchy, units and accounting signs (△ negative, − nil), with no hard-coded coordinates. Output: one Excel workbook per PDF, one sheet per statement, plus Validation and Metadata sheets.

Approach

- **Triage and routing.** A healthy text layer is used directly. A scan, or a page with no usable text, goes to OCR (YomiToku, chosen over PaddleOCR in a bake-off), after de-skewing.
- **Extraction.** Rows come from line spacing, value columns from the right edges of the amounts, hierarchy from label indent, header and period from the block above the first amount. A second strategy uses shaded row bands when they exist. Every tolerance is a fraction of a length measured on the page (word height, row pitch), so scale and layout do not matter; tested from 0.4x to 3x.
- **Validation.** Assets = liabilities + equity, totals equal their components (rounding tolerance), income-statement and cash-flow formulas from a readable YAML rule file (J-GAAP and IFRS variants). An unreadable cell is a loud failure, never a zero.
- **OCR safety net.** The accounting checks cannot see a misread label, so a page can be read twice and compared, and labels one character from a known account name are flagged. Nothing is corrected silently.
- **Layouts.** Note-number column, wrapped labels, two-panel account form, IFRS statements, page-number footers.
- **UI.** Local Streamlit page (python -m fin_table_parser ui): upload a PDF, see tables, checks and the page image, download Excel or CSV. Each table is marked ok, review (OCR), check, unverified or unsupported.

Results

What was measured	Result
Sample page vs ground truth (human-reviewed, 43 rows)	43/43 rows, 86/86 values, labels, levels, paths, types 100%; page exact
Whole company filing, 13 test PDFs (12 PDFs of one company, 1 IFRS filing of another)	all statements found; 0 failed checks (e.g. 80 passed on the filing, 119 on the 109-page annual report)
Text-only strategy vs band strategy	identical output on 28 of 28 comparable pages
OCR, YomiToku, 6 pages, clean / scan-like	cell F1 97% / 93%, about 12 s per page (PaddleOCR: 84% / 91%, about 72 s)
Wrong OCR pages caught (22 pages x 3 simulated conditions)	accounting checks alone 10 of 15; with read agreement and label check 15 of 15; 8 of 51 right pages flagged
Harder photo-like scans (blur, noise, uneven light)	about 93% of values right; 21 of 22 wrong pages flagged
Quality gates	189 tests, mypy strict and ruff clean

Limits and honest caveats

- Exact ground truth exists for one page. Other references are the parser's own output that passes every accounting check (“silver”), so a mistake shared by the text layer and the extractor would be invisible.
- Test documents are one company plus one other filing; the other layouts were tested on synthetic pages only. No real scanned PDF was available: scans are simulated (resolution, JPEG, skew, blur, noise).
- Not supported: grid-ruled pages (equity statement, notes) and statements with more than two panels. OCR can merge two-line column headers on tight layouts, so periods may be wrong on scans. A flagged cell shows where to look, not which read is right.